2026 Deloitte Digital Camp

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AI-Driven Smart Pricing and Ticketing Strategy Optimization

Team B
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The Problem :

What happens when a park loses *2.3 crore every monsoon weekend* -not because demand disappears, but because its pricing doesn't know it's raining?

The question every park operator ignores. The answer changes how an entire industry prices itself.

Pricing is still stuck in 1995 !

68%

of parks globally still use static seasonal tiers

3–4 price points per year

$2.1B

annual revenue left untapped by top-50 parks

≈$42M per park per year

43%

of visitors arrive on peak days, creating 3× queue times

crowd kills spend-per-head

0

major parks use real-time weather-conditional pricing

despite 60%+ no-shows being weather-driven

PRISM

Predictive Revenue Intelligence for Smart Multi-Venue Management

A universal demand-elasticity engine — built for theme parks, architected for any venue-based service

01 Data Dashboard & Narrative Layer

LLM-generated Morning Briefing · trend analytics · multi-audience design

02 Pricing & Product Decision Engine

Two-stage optimisation · demand forecast → constrained revenue max · segment pricing

03 Market Monitoring & PESTEL Early Warning

6 signal pipelines · anomaly detection · automated + escalation responses

PRISM doesn't ask what should we charge today. It asks what this specific visitor, in this specific context, believes is fair - and prices to that belief.

The Prism Architechture :

Five dimensions. One continuous score each. All feeding a single optimization problem

TDEI — Temporal Demand Elasticity Index

$\beta_1 \approx 0.38$

Decomposes annual demand into harmonic cycles. A continuous 0–1 score that knows how fast demand rises into a peak and when the plateau ends.

WAAP — Weather-Adjusted Attendance Probability

$\beta_2 \approx 0.24$

Converts a weather forecast into a per-ticket show-up probability. Key: an interaction term (RainProb × OutdoorFraction%) makes this calibrated to each specific park.

VSPSS — Visitor Segment Price Sensitivity Score

$\beta_3 \approx 0.21$

k-means on behavioral features identifies 4 visitor archetypes. Each has its own elasticity coefficient estimated via demand curve regression. One price fits none.

CPI — Competitive Positioning Index

$\beta_4 \approx 0.10$

Not a pricing driver — a guardrail. Keeps recommendations within a feasible corridor vs. competitors. Prevents PRISM from pricing absurdly even when internal demand supports it.

CUS — Capacity Utilization Signal

$\beta_5 \approx 0.07$

The only real-time dimension. IoT sensor data feeds a bandit agent every 15 minutes to smooth crowd distribution. Reward function penalizes congestion, not just revenue.

TDEI — Temporal Demand Elasticity Index

TDEI is predictive, not just descriptive

Most systems know Christmas is busy. TDEI knows how fast demand rises into it, how broad the peak is, and whether the shape is predictable year-over-year. It outputs all of that as one continuous number — not a calendar category.

The Formula :

$$\mathrm{TDEI}(t) = \beta_0 + \Sigma_k \left[\alpha_k \cdot \cos(2\pi kt/52) + \gamma_k \cdot \sin(2\pi kt/52)\right] + \Sigma_j \delta_j \cdot \exp(-\lambda|t - h_j|) + \varphi \cdot \mathrm{DayOfWeek}(t)$$

Holiday Decay (t, h) = exp(−λ | t−h |)

↞ λ calibrated per holiday type

k = 1..4

(k=1 annual , k=2 semi-annual , k=3 quarterly , k=4 monthly)

Output: continuous ∈ [0,1]

↠ numeric feature into XGBoost Stage 1

PRICING ZONES BY TDEI SCORE

TDEI > 0.75 — Surge pricing activates. Price floats to +35% above base. Premium window open. System defaults to Pilgrim and Planner segment pricing.

TDEI 0.35 – 0.75 — Normal pricing window. WAAP and VSPSS carry more relative weight. CPI corridor applies at standard width.

TDEI < 0.35 — Demand stimulation bundles activate. Group discounts and F&B packages fill the park profitably — not defensively.

Why Fourier, not dummies?

Dummy variables create hard step changes. Demand doesn't work that way — it rises gradually into a peak. Fourier produces smooth transitions that are both more accurate and more useful as ML features.

Holiday decay term

exp(−λ|t−h|) captures the build-up before a holiday and the drop after. λ is fitted separately for school breaks (long tail) vs. public holidays (sharp decay). This is what makes TDEI predictive, not just descriptive.

Pricing action from score

TDEI > 0.75 ↠ surge corridor (+15–35%). TDEI 0.35–0.75 ↠ standard window. TDEI < 0.35 ↠ stimulation mode, discount bundles activate. Continuous input, categorical-grade output.

Scan QR to adjust variables & see the TDEI curve respond.

WAAP —
Weather - Adjusted Attendance Probability

A 60% rain forecast doesn't mean 60% fewer visitors. It means a different visitor mix shows up.

WAAP converts the forecast into a per-ticket show-up probability — with an interaction term that makes it specific to this park's outdoor exposure.

The Formula :

$$P(\text{attend} \mid \text{weather}) = \sigma(\beta_0 + \beta_1 \cdot \text{Rain} + \beta_2 \cdot \text{TempDev} + \beta_3 \cdot \text{Wind} + \beta_4 \cdot \text{Humidity} + \beta_5 \cdot \text{Rain} \times \text{Outdoor \%})$$

AUC ≥ 0.82 · refreshed every 6 hours · park-specific coefficient

When WAAP < 0.60, PRISM doesn't just discount, it restructures the product into a Rainy Day Bundle.

where : Rainy Day Bundle = discounted entry + Indoor Priority Pass + F&B voucher.

Historical conversion : 34% upgrade rate, ¥41 incremental per visitor.

VPSS —
Visitor Segment Price Sensitivity Score

One price fits none.

Two visitors — one booked 6 weeks ago from overseas, one decided this morning. Their price elasticity is 4× apart. VSPSS uses k-means behavioral clustering to find these archetypes, then fits a demand curve to each.

A — Pilgrim $\varepsilon = -0.4$ +25–40% above base	**B — Planner** $\varepsilon = -0.9$ +5–15% + bundle
C — Opportunist $\varepsilon = -1.6$ Base or −10%	**D — Impulse Group** $\varepsilon = -2.1$ −20% + F&B upsell

Revenue-max price per segment : $p^*_k = \varepsilon_k \cdot MC / (\varepsilon_k - 1)$,

Demand + revenue curves live in PRISM Explorer.

Scan QR to adjust rain probability and temp deviation (WAAP), and see demand curve shifts per visitor segment (VSPSS).

PRISM - Interactive Model Interrogator

The pricing surface is non-linear. This is what it should look like.

Four live modules · Adjust variables · Watch pricing decisions update in real time

Scan to explore

Adjust sliders · Live pricing decisions · All 4 modules

PRISM EXPLORER

External modules (full dashboard, PESTEL, narrative layer) omitted to focus on core decision logic · Pricing surface is non-linear and cannot be fully represented statically

The Core Engine :

Two-Stage Optimization Engine

Stage 1 forecasts demand. Stage 2 takes that forecast and finds the price vector that maximizes total RevPAX — subject to six constraints encoding real business requirements, not just revenue logic.

Stage 1 — Demand Forecasting

XGBoost / LightGBM

Trained on 3 years historical data. Features: all 5 dimension scores + day-of-week + rolling 7-day average + same-week prior year. TimeSeriesSplit CV — no leakage. SHAP values explain every prediction.

Target MAPE < 8% vs. industry baseline 18–25%.

Deterministic and auditable: same inputs, same output, always.

Stage 2 — Constrained Optimization

SLSQP (Sequential Least Squares Programming)

Non-linear objective. Six inequality constraints. Resolves in < 2 seconds — enabling 15-min CUS cycles.

Output: full price vector across all time slots and visitor segments.

Market (CPI) - Prices stay within competitive corridor

Capacity - Cannot oversell beyond zone limits

Experience - Queue bounded — revenue and NPS trade-off explicit

Price stability - Adjacent slots can't jump more than Δp_max

Brand floor - Minimum price per segment — protects perceived quality

Fairness $\theta \leq 2.5$ - No segment charged > 2.5× another for equal access

The Gen-AI Architechture :

The data doesn't just display, but very skillfully informs all to know.

Every morning at 07:30, an LLM takes the optimization engine's JSON output and writes a four-paragraph plain-English briefing for the park manager. Numbers are fact-checked against source data before display. No dashboard literacy required.

PRISM | **Daily Briefing . Saturday, April 20 . 07:30 AM** **Auto-Generated**

A−	Med	18,240	¥328
Demand grade	Weather risk	Forecast visitors	Optimal base

▲ Grade A− — **92nd percentile** this week. TDEI **0.82**. School break + Sunday holiday creating long-weekend amplification effect.

Rain **58%**, 14:00–16:30. WAAP **0.64** for afternoon slots. Activate **Rainy Day Bundle at 11:30** — ¥248 + Indoor Priority Pass. Historical: **34%** upgrade rate, **¥41 incremental** per visitor.

Competitor Park X dropped 12% yesterday. CPI at +9% above market. **Hold pricing** — their discount is a TDEI 0.41 day. Matching costs ~¥380,000 in unrealized revenue today.

HOW IT WORKS

Engine outputs structured JSON

Optimization pipeline produces a machine-readable object every morning: dimension scores, forecasts, zone status, price recommendations.

LLM translates — doesn't decide

Fine-tuned model with strict rules: action-first, specific numbers only, no ML jargon, max 4 paragraphs. It translates the engine. It does not override it.

Fact-check before display

Every number in the briefing is cross-referenced against source JSON before delivery. Hallucination risk for factual claims is structurally eliminated.

The Technology Stack :

Detect market shifts before the demand data knows

PESTEL Early Warning System

Six signal pipelines. Each has its own data source, detection algorithm, and response protocol. One can fail without affecting the others.

Political

Holiday NLP ↠ TDEI recalculate

Environmental

Weather API 6h ↠ WAAP update

Social

Isolation Forest ↠ ±TDEI adj.

Technological

App audit ↠ CPI inputs update

Economic

Macro ↠ VSPSS recalibration

Legal

Regulatory feed ↠ escalate only

Deterministic and auditable:
same inputs, same output, always.

XGBoost / LightGBM

Stage 1 demand forecast · Tabular data + SHAP interpretability. Not LSTM — overkill here.

SLSQP (SciPy)

Stage 2 optimization · Non-linear objective + constraints + < 2s solve. Not deep RL.

UCB1 Bandit

CUS real-time pricing · Low-dim state space. Near-optimal at 15-min granularity. No PPO.

Fine-tuned LLM

Morning briefing generation · JSON ↠ plain English + fact-check layer. Hallucination structurally blocked.

Graph Neural Net

Phase 3 — cross-zone spend · Parks are graphs. GNN learns how pricing one zone affects adjacent spend.

Feasibility & Scalability :

Same engine. Different weight profile. Not four platforms — one architecture.

The five dimensions are universal to any venue-based, time-sensitive, multi-segment service. What changes across domains is the weight profile — not the platform, not the optimizer, not the LLM layer.

Theme Parks	Cinema Chains	Intercity Bus	Event Venues
Primary case	WAAP flips sign — rain sends people in	CPI dominates — commodity product	VSPSS critical — VIP vs GA

Stays identical :
5-dimension framework, SLSQP optimizer, LLM layer, PESTEL monitoring, API contracts, Dashboard pattern

Adapts :
Feature engineering, WAAP sign direction, Segment clusters, Price floors, Competitor targets

Impact & Implementations :

What changes. Why it holds.

+12–18%	−22%	< 8%	+8 pts
RevPAX uplift	**Peak queue time**	**Demand MAPE**	**NPS improvement**
total revenue per admission	CUS crowd redistribution	vs 18–25% industry baseline	shorter queues, fairer pricing

Phase 1 ·
~$150K · Risk: Low

Data infrastructure · TDEI model training · basic dashboard without LLM · manual price output reviewed by analyst

Phase 2 ·
~$280K add. · Risk: Medium

WAAP + VSPSS models · Stage 2 optimizer live · LLM briefing layer · PESTEL Environmental + Social · A/B pricing framework

Phase 3 ·
~$180K add. · Risk: Low

CUS real-time bandit · CPI monitoring · full PESTEL suite · GNN cross-zone spend · domain expansion pilot

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Cold-start mitigation: Parks lacking historical data enter via a Phase 0 warm-up, using a synthetic bootstrapping protocol (Gaussian copula on observed booking patterns, producing sufficient signal to initialize TDEI and VSPSS at ≥80% of trained accuracy before Phase 1 deployment.